

Practice Quiz: Learning (Mathematical Frameworks)

Name: __ Date: __

Part A: Multiple Choice

1. Parameter learning in Active Inference updates: A) The agent's beliefs about hidden states B) The concentration parameters of the A and B matrices C) The agent's physical body D) The number of observations
2. Dirichlet distributions are used for parameters because: A) They are the only distribution B) They are conjugate to the categorical distribution — posterior stays in the same family C) They are always uniform D) They represent continuous data
3. As concentration parameters grow larger: A) The agent becomes more uncertain B) The agent becomes more confident and the effective learning rate decreases C) The agent forgets everything D) The model structure changes
4. Bayesian Model Reduction (BMR) selects: A) The most complex model available B) The simplest model that explains the data well — implementing Occam's razor C) A random model D) The model with the most parameters
5. The effective learning rate decreases because: A) The brain gets tired B) Each new observation has proportionally less influence as total experience grows C) The agent stops observing D) Synapses degrade
6. Structure learning differs from parameter learning because: A) Structure learning changes what states and connections exist, not just their values B) They are the same thing C) Structure learning is faster D) Parameter learning requires no data
7. BMR is computationally cheap because: A) It uses quantum computing B) It compares model evidence analytically without re-fitting from scratch C) It ignores all data D) It only considers two models

Part B: Short Answer

1. An agent starts with $\alpha = [2, 2]$ (2 outcomes). After 8 observations of outcome 1 and 2 observations of outcome 2, compute: (a) $\alpha_{\text{posterior}}$, (b) expected probabilities, (c) the effective learning rate for the next observation.
2. Explain why the natural learning rate decay (from Dirichlet updating) is generally optimal, but give one example where it could be harmful (e.g., in a suddenly changed environment).
3. Compare parameter learning and structure learning. Give a real-life example of each. When would the brain/agent choose to restructure its model rather than just updating parameters?